

SHIMOFUSA AB, Japan

WMO: 477270

Lat: 35.799N

Lon: 140.012E

Elev: 33

StdP: 100.93

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.0	-1.0	-13.0	1.2	4.3	-11.8	1.4	4.8	9.5	8.0	8.3	7.4	1.5	310	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	33.8	26.1	32.2	25.4	31.1	25.2	26.7	31.2	26.2	30.2	25.8	29.6	4.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.1	29.5	25.1	20.3	28.7	24.8	19.9	28.5	84.3	31.7	82.0	30.5	79.8	29.7	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	7.8	6.8	DB	-4.2	36.0	1.2	1.5	-5.1	37.1	-5.8	38.0	-6.5	38.9	-7.4	40.0	(4)
(5)			WB	-5.8	28.0	1.0	0.7	-6.5	28.5	-7.1	28.9	-7.6	29.3	-8.4	29.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	4.8	5.6	8.8	14.1	18.7	21.9	25.9	27.2	23.5	18.0	12.1	7.0	(6)
(7)		DBStd	8.18	2.14	2.58	3.07	3.30	2.73	2.57	2.92	2.57	3.22	2.86	3.01	2.68	(7)
(8)		HDD10.0	464	162	125	61	6	0	0	0	0	0	0	12	99	(8)
(9)		HDD18.3	1812	420	356	294	131	28	3	0	2	41	187	351	(9)	
(10)		CDD10.0	2545	1	3	25	128	270	357	494	534	404	247	76	7	(10)
(11)		CDD18.3	852	0	0	0	4	40	110	236	276	157	29	1	0	(11)
(12)		CDH23.3	6764	0	0	0	14	176	540	2089	2754	1082	108	1	0	(12)
(13)		CDH26.7	2275	0	0	0	1	24	108	761	1066	301	13	0	0	(13)
(14)	Wind	WSAvg	3.0	2.5	2.9	3.2	3.6	3.5	3.1	3.4	3.4	3.1	2.6	2.2	2.2	(14)
(15)	Precipitation	PrecAvg	1452	66	65	106	114	130	149	132	122	200	203	105	61	(15)
(16)		PrecMax	1961	148	127	192	206	213	283	279	286	399	599	249	193	(16)
(17)		PrecMin	1041	1	9	43	42	63	44	16	14	35	26	7	2	(17)
(18)		PrecStd	209	42	34	37	46	41	52	78	80	98	134	57	39	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.2	18.0	21.0	25.1	29.2	31.8	35.2	36.0	33.0	28.2	22.2	18.1	(19)
(20)			MCWB	10.4	12.3	13.3	16.4	18.7	23.3	26.1	26.6	24.9	21.2	16.4	13.2	(20)
(21)		2%	DB	13.0	14.9	18.2	23.0	26.9	29.2	33.2	34.0	31.2	26.0	20.2	15.2	(21)
(22)			MCWB	7.6	9.8	12.0	15.7	18.5	22.3	25.8	26.4	24.8	20.4	15.4	10.1	(22)
(23)		5%	DB	11.1	12.9	16.9	21.2	25.1	27.9	32.1	32.9	29.9	24.2	19.0	13.9	(23)
(24)			MCWB	5.4	7.4	11.9	15.2	18.2	21.6	25.3	25.9	24.5	19.3	14.1	8.9	(24)
(25)		10%	DB	9.9	11.0	15.1	20.0	23.9	26.2	30.9	31.8	28.2	22.9	17.2	12.1	(25)
(26)			MCWB	4.6	5.8	10.3	14.5	17.8	21.1	25.2	25.7	23.6	18.4	13.2	7.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	14.4	15.6	20.1	22.5	25.2	27.5	27.9	26.4	23.6	18.6	15.0	(27)
(28)			MCDWB	14.3	16.9	17.8	21.7	24.9	29.4	32.9	33.6	30.2	25.9	20.7	17.1	(28)
(29)		2%	WB	8.5	10.6	14.2	17.9	20.7	24.2	26.6	27.1	25.8	21.9	16.8	11.6	(29)
(30)			MCDWB	11.3	13.6	17.1	21.3	23.8	27.1	31.3	31.6	29.1	24.8	18.9	14.3	(30)
(31)		5%	WB	6.5	8.3	12.7	16.6	19.8	23.3	26.0	26.5	25.2	20.6	15.6	9.8	(31)
(32)			MCDWB	10.0	11.0	15.8	19.7	23.1	26.0	30.3	30.6	28.3	23.2	17.8	12.4	(32)
(33)		10%	WB	5.1	6.5	10.7	15.5	18.9	22.4	25.5	26.1	24.6	19.4	14.4	8.3	(33)
(34)			MCDWB	8.4	9.8	14.1	18.7	22.4	25.1	29.5	29.9	27.5	22.0	16.5	11.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.3	9.2	9.4	9.3	8.5	7.1	7.0	7.2	6.9	7.4	8.5	9.1	(35)
(36)			MCDBR	11.6	12.1	11.6	11.1	10.6	9.4	9.1	8.9	8.1	9.1	10.2	10.9	(36)
(37)			MCWBR	7.7	8.3	7.8	6.1	4.8	4.1	3.5	3.2	3.5	5.1	6.6	7.5	(37)
(38)		5% WB	MCDWR	10.4	10.7	10.2	9.0	8.5	7.3	8.2	7.8	7.1	7.6	8.8	9.3	(38)
(39)			MCWBR	7.8	8.5	8.3	6.1	5.1	4.0	3.5	3.3	3.6	5.1	6.8	7.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.379	0.428	0.498	0.548	0.569	0.603	0.595	0.581	0.513	0.482	0.426	0.382	(40)	
(41)		taud	2.238	2.092	1.929	1.852	1.858	1.841	1.908	1.932	2.083	2.097	2.168	2.249	(41)	
(42)		Ebn at Noon	795	792	768	753	743	717	718	719	747	729	732	762	(42)	
(43)		Edh at Noon	107	139	178	203	204	207	193	185	152	137	115	100	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.73	3.23	4.05	4.74	5.10	4.54	4.91	4.93	3.82	3.01	2.48	2.38	(44)	
(45)		RadStd	0.18	0.32	0.25	0.46	0.53	0.41	0.81	0.48	0.33	0.30	0.23	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page